

DNS Policy Lab Time:

Create DNS Lists:

Create a **.txt** file with the domains that you would like to block. Save the **.txt** file on your computer to test the Domain Not Found Action Add the FQDN's to be blacklisted (**Domain Not Found**) – e.g. **yahoo.com** and **msn.com**

 DNS-List.txt - Notepad

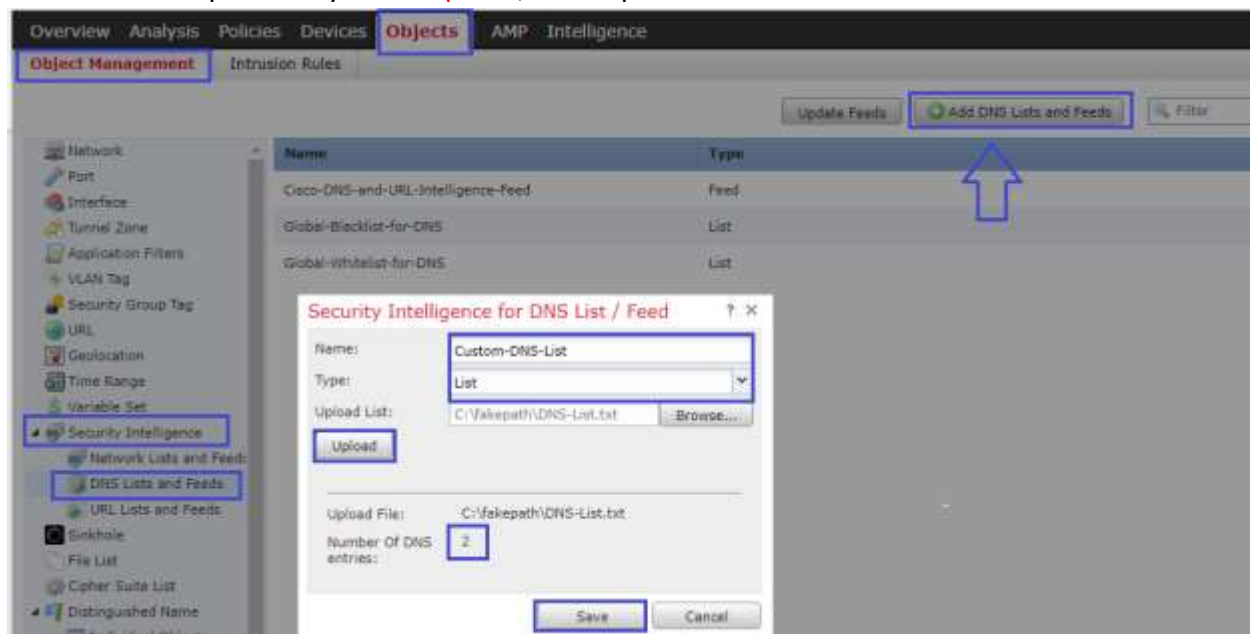
File Edit Format View Help

yahoo.com

msn.com

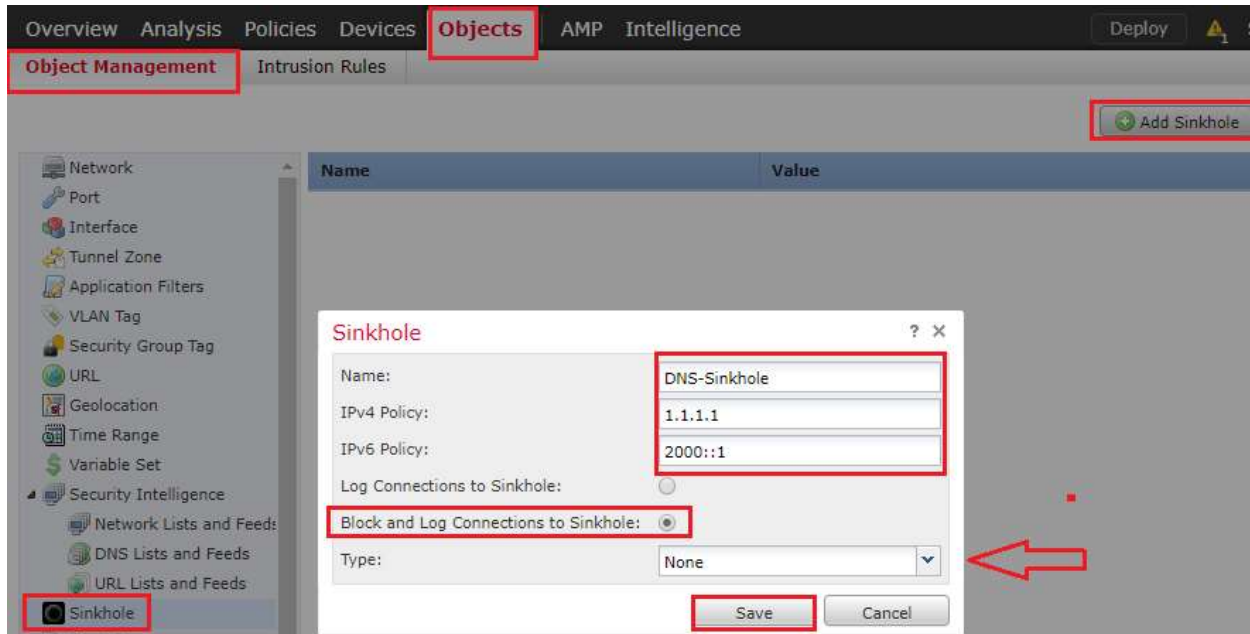
Create Custom DNS List:

Login to the FMC, Navigate to **Objects > Object Management > Security Intelligence > DNS Lists and Feeds**. Click **Add DNS Lists and Feeds**. Name the List appropriately, e.g. **Custom-DNS-List**. From the **Type**: drop-down list, select **List**. Click Browse and select the text file called **DNS-List.txt** created previously. Click **Upload**, once uploaded select **Save**.



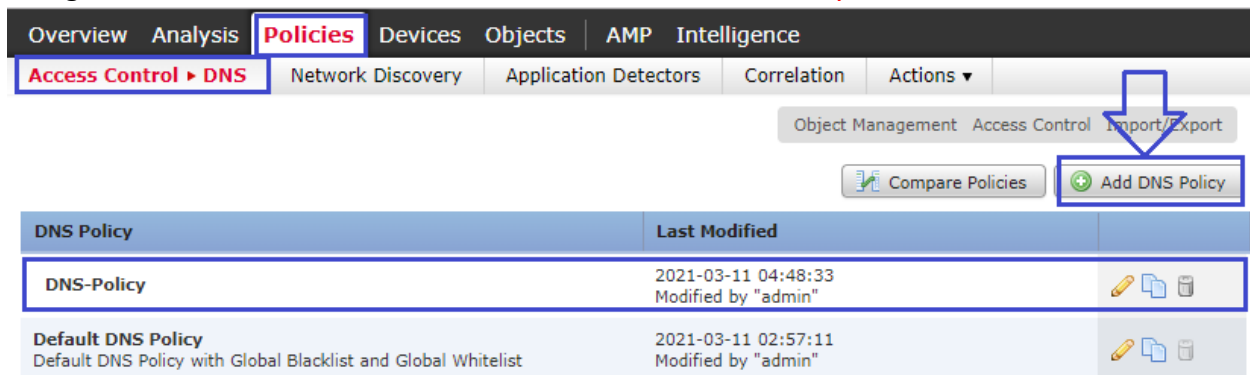
Create Sinkhole:

Navigate to **Objects > Object Management > Sinkhole**. Click Add Sinkhole. Name the Sinkhole appropriately, e.g. **DNS-Sinkhole**. Enter an IPv4 address (publicly routable e.g. 1.1.1.1 so the traffic would be routed outside of the local network). Click **Block and Log Connections to Sinkhole**. Leave Type as **None**. Click **Save**.



Create DNS Policy:

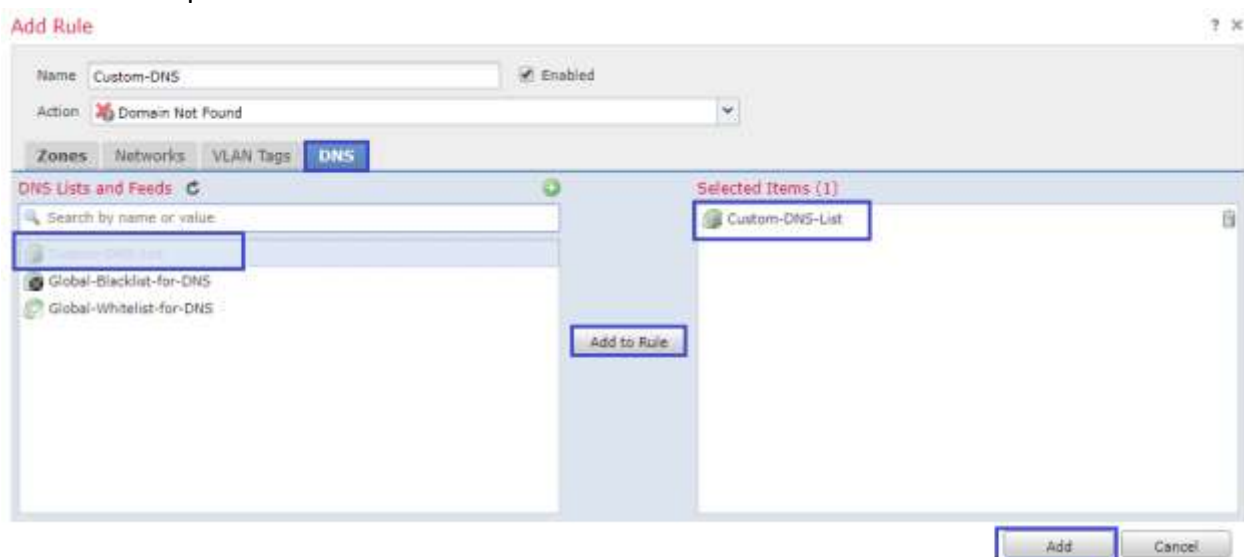
Navigate to **Policies > Access Control > DNS**. Click Add DNS Policy.



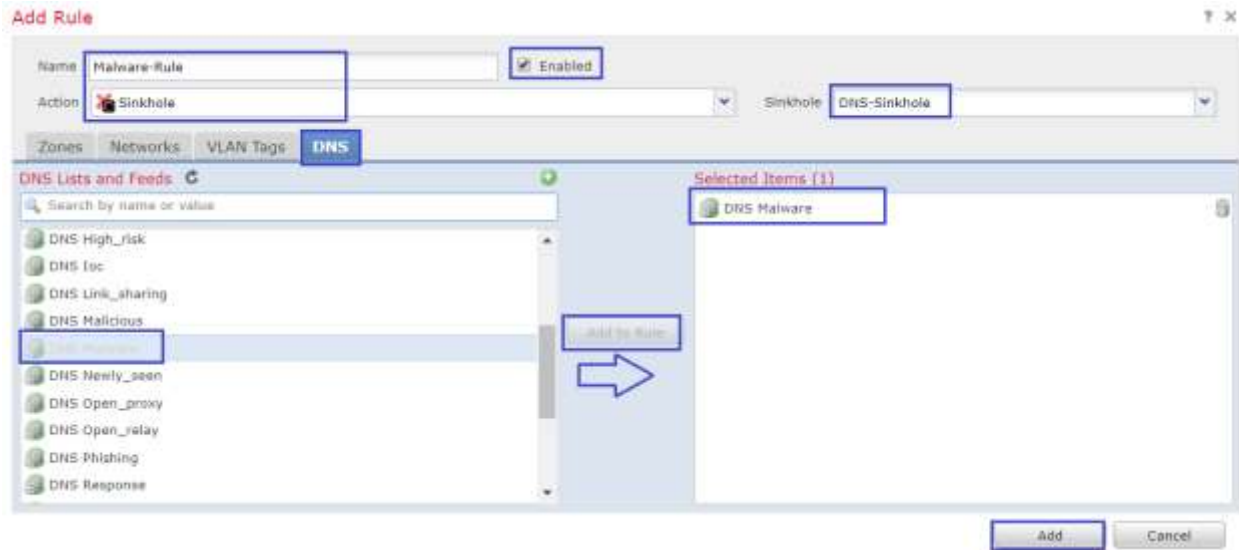
Click **Add DNS Rule**. Enter and appropriate name e.g. **Custom-DNS**. Ensure **Enabled** is checked. Select the **Action** as **Domain Not Found**. Define the Source Zone(s) as the Inside Zone.



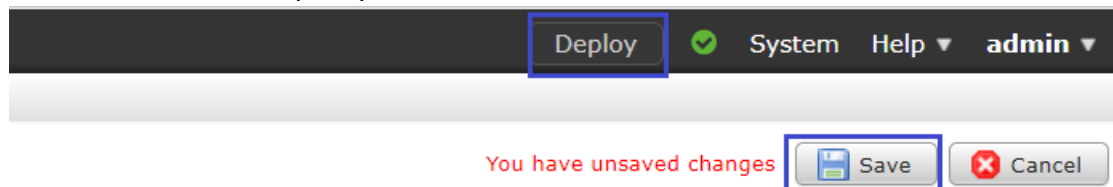
Click the DNS tab. Select the custom DNS list called **Custom-DNS-List** previously created. Click **Add** once complete. Click **Save**.



Click **Add DNS Rule**. Enter and appropriate name e.g. **Malware**. Ensure **Enabled** is checked. Select the **Action** as **Sinkhole**. A drop-down list called Sinkhole will appear, select the custom Sinkhole previously created. Define the Source Zone(s) as the Inside Zone. Click the DNS tab. Select built-in Cisco Security Intelligence feed called **DNS Malware**. Click **Add** once complete.

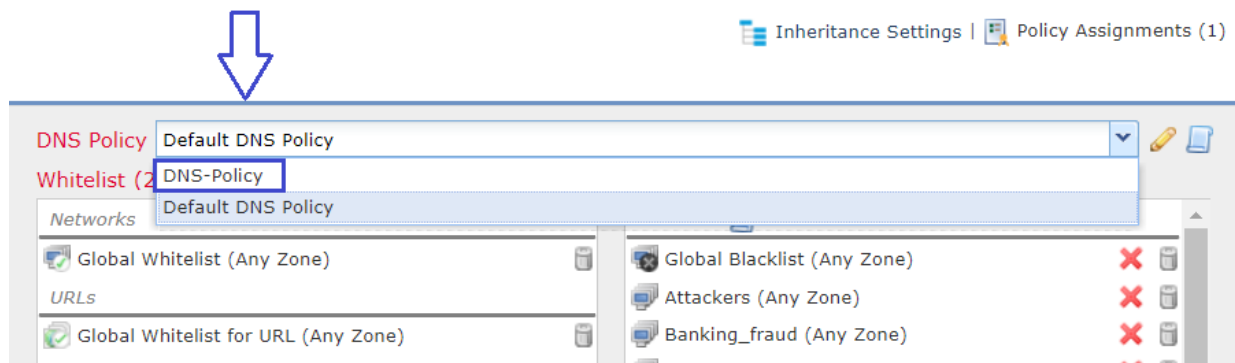


Click **Save** to save the policy.



Attach DNS Policy to Access Control Policy:

Navigate to **Policies > Access Control > Access Control**. Modify the existing Access Control Policy. Click the **Security Intelligence** tab. From the **DNS Policy** drop-down list select the DNS Policy previously created. Click **Save**.



Deploy the Policy:

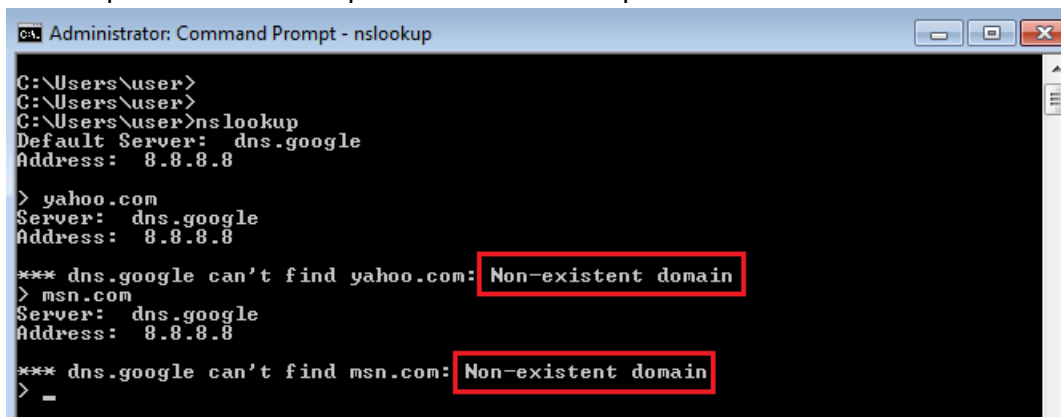
Deploy the Policy to the Cisco Firepower Threat Defense (FTD).



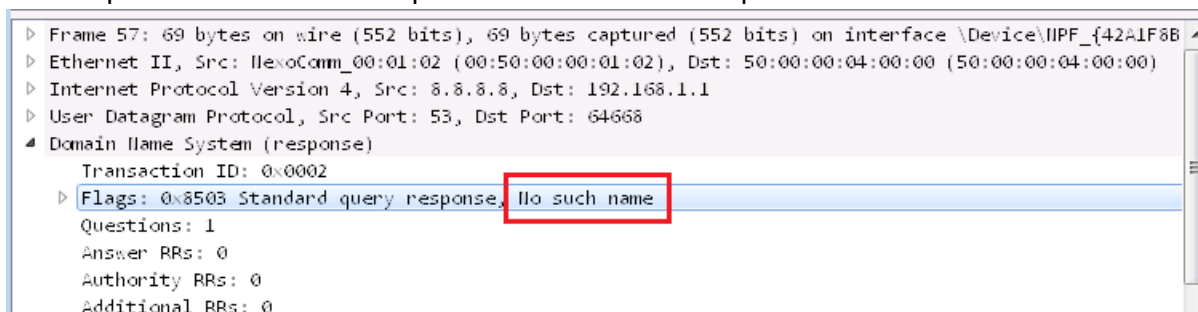
Verification:

Run Wireshark on a workstation and filter on DNS. Run Nslookup on www.yahoo.com or msn.com these domains were defined in the Domain Not Found list.

The output of the Nslookup will confirm the response “Non-existent domain”



The output of the Wireshark capture will confirm the response “no such name”.



Login to the CLI of the FTD and enter **expert mode**. Change the current directory to the Security Intelligence directory with the command **cd /var/sf/sidns_download**. List the contents of the directory using the command **ls -l**. Each file represents a DNS Security Intelligence List/Feed used when creating the DNS Policy, within each file contains thousands of domains.

```

10.1.1.10 x
root@FMC:/var/sf/sidns_download#
root@FMC:/var/sf/sidns_download#
root@FMC:/var/sf/sidns_download# cd /var/sf/sidns_download/
root@FMC:/var/sf/sidns_download# ls -l
total 433144
-rw-r--r-- 1 root root      67 Mar 12 06:24 02213098-6d94-4680-8ce8-2d081638b269
-rw-r--r-- 1 root root 772002 Mar 12 06:24 032ba433-c295-11e4-a919-d4ae5275b77b
-rw-r--r-- 1 root root    50 Mar 12 06:24 03709ed4-faab-47af-bade-4435f8da1013
-rw-r--r-- 1 www  www    48 Mar 11 09:56 0722fc92-8250-11eb-996c-a17a02ba8916
-rw-r--r-- 1 root root    53 Mar 12 06:24 1b117672-7453-478c-be31-b72e89ca2dde
-rw-r--r-- 1 root root 1715873 Mar 12 06:24 23f2a124-8278-4c03-8c9d-d28fe08b71ab
-rw-r--r-- 1 root root   327 Mar 12 06:24 2CCDA18E-DDFF-4F5C-AF9A-F00985219707
-rw-r--r-- 1 root root    56 Mar 12 06:24 2b15cb6f-a3fc-4e0e-a342-ccc5e5804576
-rw-r--r-- 1 root root    53 Mar 12 06:24 30f9e69c-d64c-479c-821d-0e4edab8348d
-rw-r--r-- 1 root root    89 Mar 12 06:24 3e2af68e-5fc8-4b1c-b5bc-b4e7cab5abcd
-rw-r--r-- 1 root root   144 Mar 12 06:24 5a0b6d6b-e2c3-436f-b4a1-48248b331d39
-rw-r--r-- 1 root root    48 Mar 12 06:24 5f8148f1-e5e4-427a-aa3b-ee1c2745d663
-rw-r--r-- 1 root root 9015846 Mar 12 06:24 60f4e2ab-d96c-44a0-bd38-830252b65259
-rw-r--r-- 1 root root    47 Mar 12 06:24 6ba968f4-7a25-4793-a2c8-7cc77f1f1074
-rw-r--r-- 1 root root    97 Mar 12 06:24 8af156ca-8020-4608-9278-01b87458fd59
-rw-r--r-- 1 root root   1058 Mar 12 06:24 8c3e31be-ca41-43c8-87cb-82a35b0f33f5
-rw-r--r-- 1 root root    46 Mar 12 06:24 937cf5e8-76d1-4ba2-a83c-475dc80c4b58
-rw-r--r-- 1 root root    53 Mar 12 06:24 A27C6AAE-8E52-4174-A81A-47C59FECd3a5
-rw-r--r-- 1 root root 403147581 Mar 12 06:22 Cisco_DNS_Intelligence_Feed

```

Use the command **grep 'Malware' <FILENAME>** or **head, tail, cat etc** with each file until the correct file has been determined. List all the current domains identified as hosting Malware.

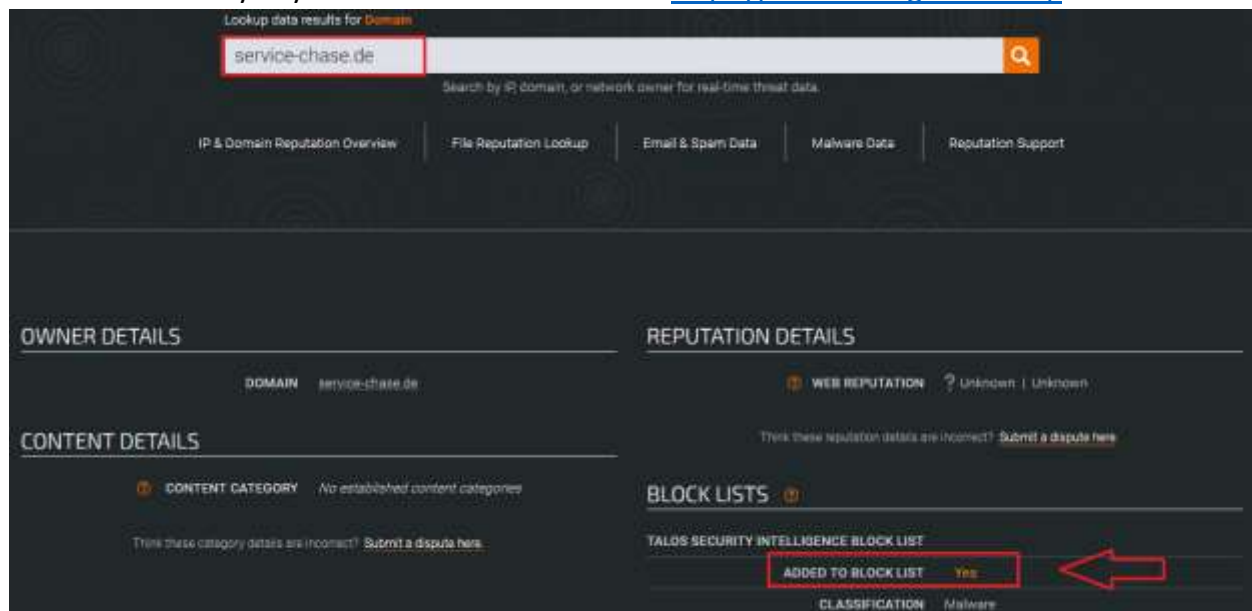
```

10.1.1.10 x
root@FMC:/var/sf/sidns_download#
root@FMC:/var/sf/sidns_download#
root@FMC:/var/sf/sidns_download# head 23f2a124-8278-4c03-8c9d-d28fe08b71ab
#Cisco DNS and URL intelligence feed: DNS Malware
uni-accweb-mouv-acadie.com
security-page.notif8.ga
netflix.errors-with-billing.com
service-chase.de
gomidbdihsbf.net
secureonline-userverification-authinformation.tk
verifysaftyacount-apl9384.cf
lopriofd.com
proevelotin.xyz
root@FMC:/var/sf/sidns_download#

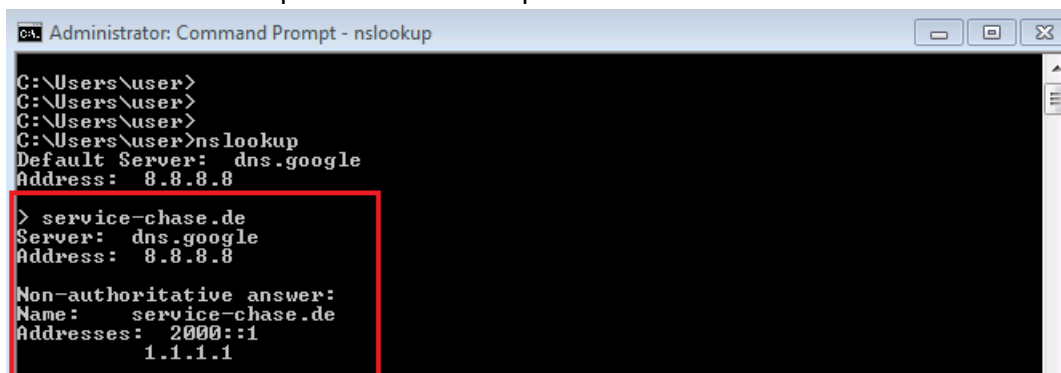
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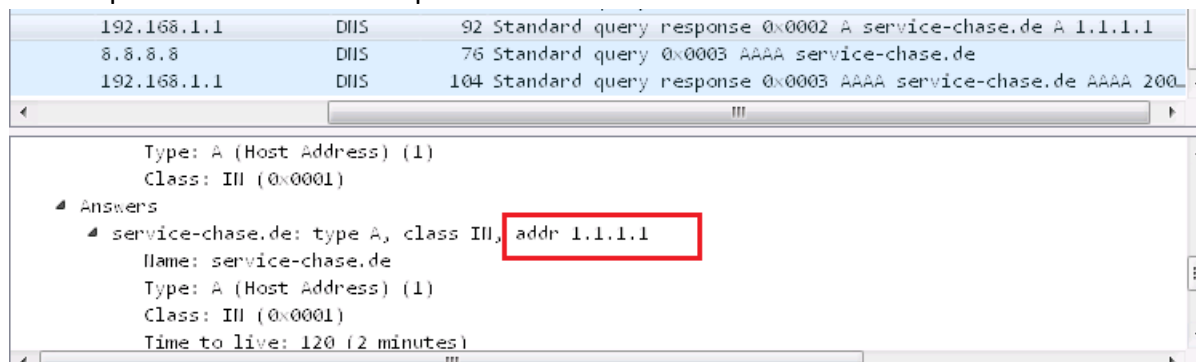
You can also verify any DNS Malware URL on Cisco <https://talosintelligence.com/>



Run Wireshark on a workstation and filter on DNS. Run Nslookup on a couple of the domains from the list. The output of the Nslookup will confirm the address of the sinkhole **1.1.1.1**



The output of the Wireshark capture will confirm the sinkhole address **1.1.1.1**



Access the **Security Intelligence Events** on the FMC. You will confirm the DNS queries matched the correct action Sinkhole or Domain Not Found.

Overview	Analysis	Policies	Devices	Objects	AMP	Intelligence
Context Explorer	Connections > Security Intelligence Events	Intrusions	Files	Hosts	Users	Vulnerabilities

Security Intelligence Events [\(switch workflow\)](#)

[Security Intelligence with Application Details](#) > Table View of Security Intelligence Events

No Search Constraints [\(Edit Search\)](#)

Jump to... ▼

	First Packet	Last Packet	Action	Reason	Initiator IP	Initiator Country	Responder IP	Responder Country	Security Intelligence Category
↓	2021-03-12 02:21:11		Sinkhole	DNS Block	192.168.1.1		8.8.8.8	USA	DNS Malware
↓	2021-03-12 02:21:11		Sinkhole	DNS Block	192.168.1.1		8.8.8.8	USA	DNS Malware
↓	2021-03-12 02:14:33		Domain Not Found	DNS Block	192.168.1.1		8.8.8.8	USA	Custom-DNS-List
↓	2021-03-12 02:14:33		Domain Not Found	DNS Block	192.168.1.1		8.8.8.8	USA	Custom-DNS-List
↓	2021-03-12 02:14:33		Domain Not Found	DNS Block	192.168.1.1		8.8.8.8	USA	Custom-DNS-List
↓	2021-03-12 02:14:33		Domain Not Found	DNS Block	192.168.1.1		8.8.8.8	USA	Custom-DNS-List

Navigate to the **Analysis > Connections > Events**. You will confirm the DNS queries matched the correct action Sinkhole or Domain Not Found.

Overview	Analysis	Policies	Devices	Objects	AMP	Intelligence
Context Explorer	Connections > Events	Intrusions	Files	Hosts	Users	Vulnerabilities

Connection Events [\(switch workflow\)](#)

[Connections with Application Details](#) > Table View of Connection Events

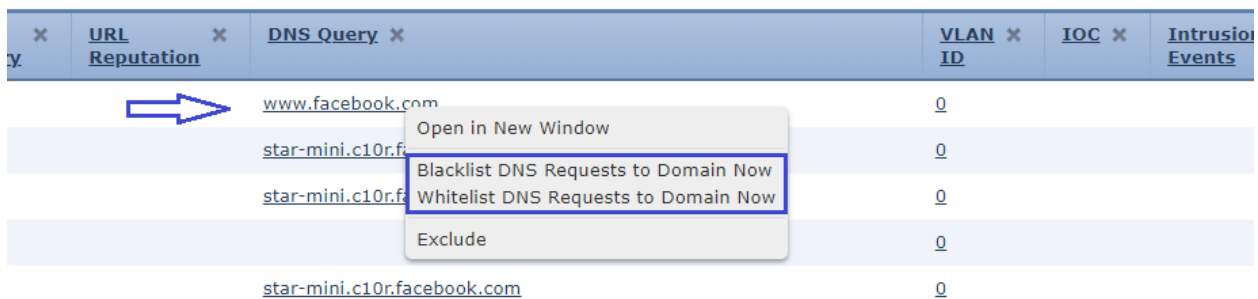
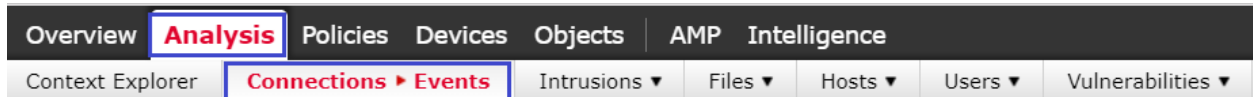
No Search Constraints [\(Edit Search\)](#)

Jump to... ▼

	First Packet	Last Packet	Action	Reason	Initiator IP	Initiator Country	Responder IP
↓	2021-03-12 02:25:47		Allow		192.168.1.1		8.8.8.8
↓	2021-03-12 02:21:11		Sinkhole	DNS Block	192.168.1.1		8.8.8.8
↓	2021-03-12 02:21:11		Sinkhole	DNS Block	192.168.1.1		8.8.8.8
↓	2021-03-12 02:21:06		Allow		192.168.1.1		8.8.8.8
↓	2021-03-12 02:14:33		Domain Not Found	DNS Block	192.168.1.1		8.8.8.8
↓	2021-03-12 02:14:33		Domain Not Found	DNS Block	192.168.1.1		8.8.8.8
↓	2021-03-12 02:14:33		Domain Not Found	DNS Block	192.168.1.1		8.8.8.8

Global DNS Whitelist/Blacklist:

To manually add a website to the Global DNS Whitelist/Blacklist. Open the Connection Events and locate the DNS Query to be listed. Locate the DNS Query. Right click and select either “Blacklist DNS Requests....” Or “Whitelist DNS Requests....”



To un-list Navigate to Objects > Object Management > Security Intelligence > DNS Lists and Feeds. Select the appropriate list “Global-Whitelist-for-DNS” or “Global-Blacklist-for-DNS” and edit.

